Eakin, H., J. Enqvist, M. Hamann, N. Methner, M. N. Sibanda, J. L. Sullivan, E. van Wyk, and G. Ziervogel. 2025. Negotiating informality and urban resilience: implications for equity. Ecology and Society 30(2):20. https://doi.org/10.5751/ES-16059-300220

Synthesis, part of a Special Feature on Enhancing Social-Ecological Resilience and Equity in Southern Africa: insights from Research and Practice

Negotiating informality and urban resilience: implications for equity

Hallie Eakin [1], *Johan Enqvist* [2,3], *Maike Hamann* [4,5], *Nadine Methner* [3], *Martha Nthambi Sibanda* [6], *Jade L. Sullivan* [7], *Ernita van Wyk* [7,8] and *Gina Ziervogel* [3,9]

ABSTRACT. Informality is a distinguishing characteristic of cities in the Global South and is strongly associated with urban inequality. Yet, in pursuing resilience, urban resilience strategies and planning have yet to grapple with the role of informality in social-ecological dynamics, resulting in incomplete representations of the reality of these cities' socioeconomic and demographic diversity. Neglect of informality has significant, but uncertain, implications for equity in resilience planning. In this paper, we conceptualize the complex, dynamic urban systems in southern Africa as emergent from the interdependent interactions between formally recognized and so-called "informal" institutions, economic activities, and social-ecological processes and entities. These interactions generate feedback and emergent outcomes locally and at the scale of the broader urban system, with complex implications for urban resilience, equity, and sustainability. We explore the role of informality in urban resilience in relation to two cases of urban environmental crises: drought in Cape Town, South Africa, and flooding in Lilongwe, Malawi. The cases illustrate how managing resilience at one spatial or temporal scale can mask or generate inequitable outcomes at other scales. The role of informality and its linkages need to be acknowledged for informality to be better incorporated into urban resilience planning, as recognition is often the first step to confronting legal and normative barriers and significant power asymmetries. Informality is a malleable social-political construct, and the actors who control its definition have significant influence over the distribution of rights, responsibilities, and resilience in urban systems. Any strategy to improve social equity in urban resilience planning therefore must address the asymmetries in power that characterize the informal/formal divide. Formally recognized organizations that can legitimately bridge informal and formal spaces play key roles in enhancing procedural, and thus distributive, justice outcomes, as well as in creating the collective capacity to address rapid urban change in the Global South.

Key Words: *climate adaptation; environmental justice; Global South; social-ecological systems; Sub-Saharan Africa; urban governance*

INTRODUCTION

Efforts to build urban resilience in southern Africa have yet to grapple with a distinguishing characteristic of the region's cities: informality. Strongly associated with inequality, informality is prevalent in institutional, spatial, political-economic, and social-ecological interactions (Banks et al. 2020) and challenges resilience work both conceptually and in practice (Parthasarathy 2015, Nagendra et al. 2018). On the one hand, informality is often perceived as a scourge that undermines resilience and needs to be "regularized" into formal management schemes (Ohnsorge and Yu 2022). On the other, informal socioeconomic activity can be a source of grassroots innovation and social resilience when formal institutions are inadequate or overwhelmed (Kumar and Bhaduri 2014). Data on informality are often lacking, making accounting for it in urban planning difficult (Tellman et al. 2021). Nevertheless, failure to do so may undermine efforts to ensure equity and justice in urban resilience.

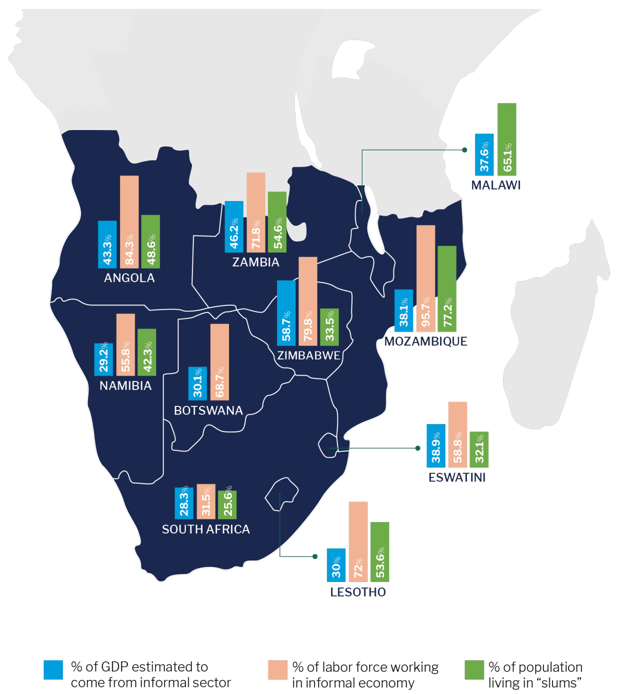
Resilience generally refers to a system's capacity to absorb, adapt to, or transform in the face of disturbance without losing key properties (Johannessen and Wamsler 2017, Elmqvist et al. 2019). Urban policies often treat resilience as a normative goal or desired state linked to sustainability (Meerow et al. 2016). As urban resilience and climate adaptation efforts advance to consider impacts on equity (Meerow et al. 2019) and justice (Bulkeley et al. 2014), resilience scholars increasingly recognize the value-laden nature of urban resilience-building (Elmqvist et al. 2019, McPhearson et al. 2021), asking: resilience for whom, what, when, where, and why? (Meerow and Newell 2016).

In much of the Global South, and increasingly in the Global North, these questions require attention to informality. Urbanization in the African continent is likely to occur faster than on any other continent (United Nations 2018). The expansion of irregular or informal settlements and economic activities are already noted as both an implication of climatic change and a concern in climate vulnerability (Brown et al. 2014, Intergovernmental Panel on Climate Change [IPCC] 2023). The complexity and multilevel nature of urban social-ecological systems (Elmqvist et al. 2019) mean that people, places, processes, and structures designated as informal will experience shocks and stressors differently than those designated as "formal." Efforts to promote urban resilience must therefore draw on context-specific knowledge about the nature and function of informality (e.g., Fig. 1) to minimize the risk of unexpected and inequitable outcomes. Such knowledge can enhance efforts across the urban Global South where informality is prevalent (Brown et al. 2014, Taylor and Peter 2014), making it critical for resilience scholars, urban practitioners, and the transdisciplinary partnerships where they collaborate to enhance equitable outcomes in urban spaces.

This paper posits that achieving greater equity in pursuit of urban resilience requires engaging with informality in its diverse manifestations. We combine a literature synthesis on informality, resilience, and equity in the Global South, a brief review of urban resilience policies, co-author experience from research and practice in the southern African region, and resilience lessons drawn from illustrative cases on flooding in Lilongwe and drought in Cape Town. Such examples of rapid system disturbance help

[1]School of Sustainability, Arizona State University, [2]Stockholm Resilience Centre, Stockholm University, [3]African Climate and Development Initiative, University of Cape Town, [4]Centre for Sustainability Transitions, Stellenbosch University, [5]Centre for Geography & Environmental Science, University of Exeter, [6]Slum Dwellers International, South Africa, [7]ICLEI - Local Governments for Sustainability, Africa Secretariat, [8]Institute for Coastal and Marine Research, Nelson Mandela University, [9]Department of Environmental and Geographical Science, University of Cape Town

Fig. 1. The level of informality in the economy (MIMiC), labor force (International Labor Organization [ILO] 2023), and settlements (World Bank 2023) across ten southern African countries.



to "make the invisible visible" and allow for examining how informality both underpins and undermines urban system resilience. We show the day-to-day contributions of informality in so-called "normal" times become visible and even essential in moments of turbulence, before formal institutions and actors take over responses, regulate processes, and convey formal rights and responsibilities. Lastly, we reflect on the implications for social equity of expanding urban resilience strategies to encompass informality.

INFORMALITY

Informality is a relational term, impossible to define without referring to what is codified, legislated, and regulated by "formal" institutions and governance arrangements, and to prevailing social norms. Any definition of what constitutes the "informal" is therefore highly contextual. Informal activities and spaces are typically unregulated by the state (Auerbach et al. 2018). Informality refers to a range of unwritten rules, social norms, and activities that circumvent or are beyond the law, and to practices that get things done where formal institutions are absent, ineffective, or irrelevant. Although often stigmatized through linkages to rule-breaking, corruption, and illicit activities, many informal practices emerge out of people's desire to help family, friends, and neighbors (Global Informality Project 2024). From a more critical perspective, Roy (2005) argues that informality in urban contexts is not a "state of exception" to formal order but rather a "mode of urbanization"—part of the way the politics of urban development occurs. Informality is thus socially constructed, and defining what is or is not "formal" wields considerable power (Roy 2005, 2009). Informality is a function of which specific authorities are considered legitimate (and thus "formal"), how that legitimacy is conferred, by whom, and through what procedures (McClymont and Sheppard 2020). Problematically, in much of international development discourse, informality continues to be reduced to an (undesirable) anomaly, an unfortunate condition where something is lacking or has failed, with the formal being the "norm" and the preferred end-state of social dynamics (Kooy 2014). In reality the formal–informal is more of a continuum than a binary, where some "unregulated" relations may be considered completely legitimate (e.g., the direct hire of a domestic worker), and others considered illegal (e.g., an unregistered miner of minerals in an abandoned mine).

In southern Africa and the Global South more broadly, unregulated, unregistered, or otherwise unrecognized practices are the norm in terms of the social-spatial prevalence and temporal dynamism of cities (Chapman and Sasman 2012, Charmes 2019). Informal labor, economies, and settlements are common and tend to persist, uncodified, undefined, and unrecognized in parallel and interdependent with the formal (Roy 2005, Ahlers et al. 2014, Kooy 2014, Banks et al. 2020, Sheppard et al. 2020). In the Southern African region alone, informal labor force estimates range from 31% to over 95%, with 28% to 58% of GDP derived from the informal sector (Fig. 1). These figures reflect significant challenges to growth and employment in formally regulated non-agricultural activities in countries such as Angola, Mozambique, and Zimbabwe, and the continued importance of semi-subsistence agriculture across the region, the majority of which is considered part of the informal economy. As urbanization has increased across the region in the last few decades, urban populations now exceed rural ones in Angola, Botswana, Namibia, and South Africa, putting additional pressure on housing stock. Informal practices could be considered the social-ecological foundation from which formal governance carves out spaces of political, economic, and spatial authority, usually under norms of capitalist expansion (Sheppard et al. 2020). Deciding which activities, spaces, and practices become publicly recognized and legitimized is a matter of state strategy, as it influences how and when the state can be seen as accomplishing public sector goals and services (Roy 2005, McFarlane 2012, Tellman et al. 2021).

Whether understood as a mode of urbanization or state of exception, informality entails risks for the activities and actors that are given the label. Further, the designation of "informal" implies significant inequities in the distribution of the burdens and benefits of urban life, and often a lack of recognition and representation in formal governance procedures (Sarmiento and Tilly 2018). Lacking access to formal urban services of water, energy, transport, finance, and housing, populations designated informal often bear disproportionate risks from the pursuit of unregulated livelihood activities and environmental exposure (UN-Habitat 2015, Sandoval and Sarmiento 2020). Lacking access to the protections afforded populations living in formal locations may force populations of informal status to rely on exploitative and violent relations to meet basic needs (Wood 2003, Peters et al. 2022). They are more often subject to the urban externalities of municipal waste, industrial contaminants, environmental hazards, organized crime, and exploitation (Wood

2003, Brown et al. 2014, Banks et al. 2020). Climate change is exacerbating this disproportionate exposure (Dodman and Satterthwaite 2008, Revi et al. 2022) and presenting new challenges for informality (Finn and Cobbinah 2022). Women and undocumented immigrants often are among the most vulnerable to these hazards, as intersectional discrimination excludes them from both formal and informal spaces (Brown et al. 2014). High rates of un- and underemployment among populations of informal status can thwart young people's entrepreneurial energy by limiting access to educational resources and opportunities in the formal economy, pushing many into illicit activities (Losch 2016, Darvas et al. 2017).

Yet, so-called informal places and practices can also display considerable innovative, problem-solving, and entrepreneurial capacities (Daniels 2010, Akinola 2016, Enqvist et al. 2020). Informal activities can produce significant benefits for people involved in them, for their immediate communities (e.g., economic and livelihood enhancement) and for the cities where they occur. For example, informally governed economic and social networks help provide key municipal services, such as waste management and food and water provision (Ahlers et al. 2014, Battersby et al. 2016, Simatele et al. 2017, Chigwenya and Wadzanai 2020, Sheppard et al. 2020). Institutional scholars have long recognized the important resource accessing and management roles and functions of informal institutions, rules, and norms (Whaley 2018, Tellman et al. 2021).

Despite a potential for constructive social and technical innovations from informal places and activities, the complex relationships inherent in the formal–informal urban system creates uncertainty for policy and planning. In many global and local policy documents on urban resilience and sustainability planning (Box 1), informality is predominantly treated as a development problem and challenge. If such policies were based on a more comprehensive understanding of the systemic role informality plays in urban systems, they could play an important role in promoting norms and practices that recognize informality.

Box 1:

Global and local policy review on informality

(See Appendix for further details.)

Global policy on informality

Based on their global significance in addressing social-ecological resilience, we reviewed nine international policy and science consensus documents pertaining to urban sustainability and urban nature and biodiversity:

Although most documents foreground justice and equity as essential for urban resilience, only five of the nine documents addressed informality explicitly.

Where mentioned, informality typically referred to informal settlement (land use and the built environment) or informal economic activity.

In most cases, informality was associated with disproportionate vulnerability to climate-related shocks and stressors, stemming from lack of basic services, housing, insecure land tenure, and failures in social inclusion and human rights.

The majority of documents advocate for upgrading and "regularizing" informality to address these challenges.

A notable exception in global policy was the Cities Biodiversity Outlook (2013), in which informality was presented as both a challenge, because conventional regulatory measures cannot direct informal systems effectively, and an opportunity, because informality may hold benefits, such as being able to respond rapidly and flexibly in the face of change in the urban landscape. The Outlook also recognizes the growing influence of informal groups in urban decision-making and that they strengthen the governance capacity of resource constrained local public agencies.

Urban policy on informality

Across the African urban context, a review of existing urban resilience policy documents associated with six cities in which informality plays a strong role revealed a diversity of approaches to the role and function of informality in resilience. In almost all cases, informality was presented as a challenge for urban authorities, particularly in those cases where the size of the informal sector and settlement area represented a large portion of the urban population and economy. Accra's urban resilience plan, for example, emphasized the need for collaborative strategies, acknowledging the role of informality as part of the urban fabric. In Addis Ababa, informality is presented as overwhelming urban capacities, leading to policies that support grassroots efforts to address the issues informal settlements face. Despite progressive discussions in some resilience documents, the overall view is still that informality is an impediment or part of the economy that needs to be re-organized and aligned to be included in plans for the city. The idea of leaving the informal sector unregulated is seen as a barrier to progress, and a major contribution to the vulnerability of urban people when faced with shocks.

Building on the reviewed literature, we view "the informal" not merely as an absence of, but co-constituted by "formal" processes, places, and outcomes. The informal designation is a product of power relations in urban systems and thus a social and political construction. Nevertheless, it is also undeniably a material and social reality entangling many—likely a majority—in southern Africa's urban systems. As cities in this region and beyond seek to develop resilience strategies and plans in the face of growing internal constraints (e.g., population growth, finance limitations, rapid urbanization) and exogenous challenges (e.g., climate change, refugee crises), the role and reality of informality need recognition as a key influence on the resilience of urban systems (Brown et al. 2014).

URBAN RESILIENCE, JUSTICE, AND EQUITY

In resilience work, equity is usually considered either a necessary functional component of resilience (e.g., Cote and Nightingale 2011, Matin et al. 2018) or a desired and normative addition to what resilience should be (e.g., Garcia et al. 2022). Fitzgibbons and Mitchell (2019) caution that despite critical scholars' concern that failing to address equity and justice undermines resilience goals, there is limited empirical evidence to test these claims. Here we embrace both arguments, positing that a focus on informality illustrates that equity is indeed functionally critical for achieving a more resilient urban system, and that equity is a normative goal for urban system resilience that demands attention to informality.

Environmental justice research typically entails three dimensions of justice: the allocation of services, opportunities, resources, and associated costs (distributional justice); the nature and inclusiveness of participation in decision-making processes (procedural justice); and the acknowledgment of and respect for different groups, identities, and societal status resulting from historic injustice (recognitional justice; Schlosberg 2004, 2007). For urban adaptation governance, Bulkeley et al. (2014) extend their analysis to also include rights and responsibilities, and further highlight recognition as foundational for the pursuit of equitable responsibility, distribution, and governance procedures. Like justice, equity has also been described as having distributional, procedural, and recognitional dimensions (Pascual et al. 2014, Leach et al. 2018). More specifically, equity refers to positions of differential access to resources, capacities, and power among populations of interest (Matin et al. 2018). Attention to equity thus helps promote justice; circumstances of injustice create and perpetuate inequities.

Recognitional justice is particularly relevant for the analysis of informality, given that informality almost by definition misrecognizes specific populations, activities, and places by labeling them as illegitimate, illegal, or lacking rights. Recognition "brings into focus the ways in which inequalities are created and sustained by the same social, political and economic processes which determine what 'fairness' means" (Bulkeley et al. 2014:33). With a critical role in making the invisible visible, recognition can expand and more accurately define the scope and scale of urban system dynamics for resilience work. It is critical for negotiating and establishing more equitable and just procedures, resource distribution, and associated rights and responsibilities, thus helping make citizen-state social contract arrangements explicit (Schlosberg 2004, Bulkeley et al. 2014). Grappling with contentious issues of recognition, distribution, responsibilities, and procedure underscores the political nature of resilience work (Meerow et al. 2019, Eakin et al. 2022).

Although equity and justice are chronic concerns, we posit that they become more critical during periods of significant and rapid system change. Structural vulnerabilities (e.g., food insecurity, unaffordable housing, or inaccessible public services) that are masked as tolerated inequities in so-called normal times can become acutely visible in moments of crisis (Sultana 2021). They can also reveal how informal places, processes, and activities have already adapted to conditions of precarity. This "adaptedness," though often coming at a very high cost to those undertaking such adaptations, is what helps keep informality invisible and out of sight during so-called normal times (Eakin et al. 2016). Populations, activities, and places designated as informal in normal times can be further marginalized in moments of crisis, despite often being on the frontlines of adverse effects (Peters et al. 2022). Furthermore, disruptions can exacerbate equity issues by incentivizing people to act in their own self-interest without having time to consider broader implications (Simpson et al. 2019a, Dade et al. 2022). We posit that while disturbance may create windows of opportunity for new urban governance arrangements, recognition is key to allowing a renegotiation of procedures as well as of distributions of rights and responsibilities in support of enhanced urban equity. An ability to recognize how urban development and intervention (re)produce forms of social, political, and economic inequality thereby becomes central to developing new social contracts that enhance resilience at multiple levels.

CASE STUDIES

To illustrate how systemic crises can reveal informality's various functions and implications for urban resilience, we examine two cases in more detail. First is the multiyear drought that struck South Africa's second largest city. Despite having a relatively well-resourced municipality, Cape Town is characterized by extreme levels of socioeconomic inequality, where affluent households could adapt through informal groundwater sourcing, in turn impacting equity and resilience at the city-scale. The second case illustrates how informality shapes resilience to repeated flood events in Malawi's capital Lilongwe, where rapid urbanization has overwhelmed municipal service provisioning capacities. During flooding, the populations, settlements, and activities labeled informal play key roles in altering the city's exposure and resilience. In addition to these case studies, our working group has also produced a blog post, an insights brief, two posters, an infographic, and a cartoon to illustrate and communicate our message to a diverse, non-academic audience. These are available at https://www.globalresiliencepartnership.org/what-we-do/knowledge/south-to-south-resilience-academy/southern-african-resilience-academy/building-equitable-resilience-in-southern-africa/informality-and-equitable-urban-resilience/.

Cape Town

Informality, resilience, and equity came to the fore in Cape Town's 2015–2018 drought, as the city was almost completely reliant on surface water sources to serve its economically and demographically diverse population (Ziervogel 2019). Efforts to address highly inequitable Apartheid-era water access along racial lines have had results, but entrenched economic inequality impedes progress (Enqvist and Ziervogel 2019). Prior to the drought, only 4% of the municipal supply was consumed by the 15% of Cape Town's residents who live in informal settlement "townships" (almost exclusively black African and colored: a term that, while controversial in some parts in the world, is still used in South Africa's official census in reference to one of the country's four main population groups) (Department of Water and Sanitation 2018). The City provided eligible poor households with 350L of water/day for free as part of its constitutional obligations, but populations in informal settlements often lacked connections within their homes and instead relied on communal taps or more ad-hoc water delivery services. Furthermore, many household connections had access restricted by municipal "water management devices," designed to cap use to 350L and keep households from going into debt, and to help detect leaks (Mahlanza et al. 2016). In contrast, municipal water was readily accessible in more affluent and largely white neighborhoods, and, prior to drought restrictions, was often used to water lush gardens, fill pools, and other non-essential uses.

From the outset, households' drought resilience thus varied greatly. Wealthier households could meet tighter water restrictions and mitigate tariff hikes by limiting non-essential water use without endangering health or sanitation. For residents in informal settlements, on the other hand, water was typically used for cooking, washing, and cleaning, which meant that higher tariffs forced many households to choose between higher water costs or risks to basic safety and survival (Robins 2019). Their problems were exacerbated as low-income households in formally constituted areas often have "backyard dwellers," i.e., people living informally on the same property, who were essentially invisible in the drought response policy process (Ziervogel 2019). As water restrictions were set at the household rather than individual level, such dwelling arrangements often drove even essential water use above the 350L cut-off point.

The city did recognize the high vulnerability of populations in informal settlements, their right to sufficient water, and its responsibility to guarantee such rights. Communal standpipes in informal settlements were therefore not turned off during drought restrictions. Nevertheless, the interdependence of water and livelihoods in the townships was often not sufficiently appreciated by the public. During rising media scrutiny of water use, wasteful behavior was called out both at informal car washes using communal taps and in affluent areas when residents watered their lawns (Jack et al. 2019, Robins 2019). However, the expectation of equal reductions in water use from all, without regard for loss of livelihoods, created inequitable outcomes in terms of household resilience.

As the drought continued, many wealthy households responded by seeking alternative, unregulated (and thus informal) water supplies, namely, drilling boreholes and wellpoints and installing rainwater tanks and graywater systems (Simpson et al. 2019a). The number of borehole sales increased from 1500 in 2016 to 26,000 by early 2019 (Simpson et al. 2019a), but because groundwater management is a national government mandate, the city struggled to enforce a request to register household boreholes. A World Wildlife Fund [WWF] study found that 90% of boreholes in the affluent Newlands suburb were not registered in any city or national database (WWF 2020).

These informal supply strategies had a complex impact on resilience: they helped reduce strain on the municipal supply in the short term but also produced negative externalities (Simpson et al. 2020). For example, backflow from graywater systems risked contaminating the city's water mains, and unchecked groundwater use could impact longer term water availability for ecosystems and the wider community. In addition, privatized water supply also had serious indirect equity implications. This was a result of the city's progressive water tariff system, where high-volume users pay a higher rate to cross-subsidize access for low-income households and users in informal settlements. Because of this, wealthy households' shift to informal water provisioning strategies created a loss in revenue that was disproportionate to the water saved, posing a risk for the resilience of the city's water systems—especially its capacity to ensure water security for the most vulnerable (Simpson et al. 2019b).

Post-drought, the city's new water strategy is taking steps to recognize existing inequities by, for example, acknowledging backyard dwellers' rights to water, and the specific need to ensure water access in informal settlements (City of Cape Town 2019). It also recognizes the need to better understand the state of groundwater resources, and what local and national governance responsibilities regulate access and sustainable use. The city has also implemented a fixed cost for pipe connection to recuperate some of the revenue lost because of households' increased reliance on alternative (informal) sources such as boreholes and rainwater harvesting systems for non-drinking purposes.

Furthermore, the city has made efforts to engage in dialogue with civil society to address service delivery challenges in low-income neighborhoods and informal settlements. Historically, such attempts have often been one-sided and constrained by mistrust, but recent efforts have shown that bridging organizations and actors such as local non-governmental organizations (NGOs) or academics can help create legitimacy and focus dialogues, for instance by working with residents in knowledge co-production to strengthen their credibility as partners (Ziervogel et al. 2021). A different kind of bridging between the informal and formal can be seen in a "citizen science" project initiated by WWF in partnership with private consultancies, inviting private borehole owners to help monitor groundwater levels (WWF 2020). Although the WWF strategy does not attempt to "regularize" these users, it helps draw attention to their potential implications for urban resilience and equity. These examples are still the exception, however, and there has been limited success in institutionalizing them as a way to address procedural justice.

Lilongwe

Between 2017 and 2019, a series of severe floods and related disasters took place in several of Malawi's riparian cities, including the capital, Lilongwe (Kita 2017, Makuwira 2022). Although the capital is highly dependent on the Lilongwe River, rapid urbanization has made the city highly vulnerable to floods that disrupt living conditions and livelihoods (World Bank 2019), damage infrastructure, and threaten human lives (Global Center on Adaptation 2022). Vulnerability is particularly acute in burgeoning informal settlements and communities, which have high levels of poverty and limited access to the formal system of land allocation, and hence often settle on riverbanks. Flood risk is likely to persist into the future as climate change makes rainfall events more intense (World Bank 2019, African Cities Research Consortium [ACRC] 2021).

Lilongwe's population of 1.2 million is growing by 4%–5% annually (ICLEI 2020, World Population Review 2023), and authorities struggle to effectively provide basic services, such as healthcare, housing, education, waste collection, and disaster response (Adams 2017, Makuwira 2022). Inadequate infrastructure leaves many with poor sanitation and water drainage, particularly exposing informal settlements to water-borne diseases such as diarrhea and cholera (ACRC 2021). Municipal water and electricity are provided only to citizens who pay for these services (UN-Habitat 2011). This means that although Lilongwe's government has extended pipes into informal settlements, many poor people cannot afford the actual water, instead relying on community standpipes, private wells, boreholes, springs, and streams (Alda-Vidal et al. 2017, Mitlin and Walnycki 2019).

Informal economic activities offer a means of survival under highly precarious conditions. A majority of Lilongwe's vendors trade illegally outside of the council's designated areas for trade (Budlender and Gondwe 2019). Informal trade plays a key role in providing food security and income, strengthening individual, household, and community resilience (Budlender and Gondwe 2019); however, it also generates a significant amount of waste. Because formal waste removal is limited to commercial sites (UN-Habitat 2011, Lilongwe Water Board 2021), much waste is dumped along Lilongwe's roadsides and in the river, exacerbating flood risk and affecting the river's ecology. Officials see informal traders as exacerbating city congestion and competing with formal land use planning (Budlender and Gondwe 2019, ICLEI 2020), and strained relations often undermine problem-solving.

Small-scale and artisanal sand mining is also practiced in and along rivers, for use in both informal and formal settlement construction. It is illegal, but plays an important role in poverty reduction and is a source of income for both men and women lacking formal employment (Makuwira 2022). Sand miners sell sand to intermediaries who transport the product to markets, creating a value chain (Malebana 2021). However, sand mining bears its own inequities as established miners control the earning power of less established and temporary miners (Fandamu and Fandamu 2018). Sand mining also alters river morphology, exacerbating flood risks both on-site and in downstream riparian areas (Ramkumar et al. 2015, Makuwira 2022).

Informality is also implicated in flood responses. Community-based flood risk management plays a critical role in Malawi, facilitated by donor funds and implemented by local NGOs (Šakić Trogrlić et al. 2017). For decades, this system has mobilized local knowledge and skills to stimulate cost-effective flood risk preparedness and mitigation. Activities include reforestation, installing gauges to track water levels, social innovations such as disseminating warning messages to cellphones, search and rescue protocols, and distribution of relief items. However, since government policies rarely incorporate these community-level systems, the burden of disaster response remains with communities themselves while underlying drivers of vulnerability remain unaddressed. This long-practiced, externally supported, local-level resilience thus alleviates the city administration's obligation to address disaster-related assistance. Ironically, external assistance by NGOs to local communities may entrench these communities' reliance on their own (informal) risk management strategies, and exacerbate the lack of recognition of their rights by the formal governance system.

As in Cape Town, bridging organizations can help to forge more constructive relationships between formal and informal actors and processes. One example is the Malawi Alliance, a Slum Dwellers International Affiliate made up of the Centre for Community Organization and Development and the Federation of the Rural and Urban Poor, a network of community-based organizations. These partners work together to promote public/private partnerships in efforts to enhance service delivery in informal settlements, in particular leveraging land tenure for housing, sanitation projects, and the creation of climate risk awareness and accountability in community-led monitoring and evaluation. The federation has historically worked closely with city officials (on early warning systems, emergency response, and the rebuilding of informal communities), despite consisting of informally organized savings groups. NGOs have also been able to mediate relations among informal market traders and local government authorities to address the problem of solid waste contamination in the Lilongwe River. An international network organization, ICLEI, was able to broker improved relations between traders operating informally on the Lilongwe River banks and the Lilongwe municipality (ICLEI 2020). As a result, city officials have recognized the informal market's governance structures and worked with informal market leaders to co-design a program for waste composting and a plan for rehabilitating the affected riparian area (ICLEI 2020). These examples illustrate how NGOs can facilitate knowledge flows and broker relations to bridge informal and formal actors and systems (Frantzeskaki et al. 2019), something that Lilongwe City Council has explicitly acknowledged with regard to the Malawi Alliance (Global Center on Adaptation 2022:27). (Several authors in this paper have played a part in these types of brokering efforts, both in Cape Town and Lilongwe.)

DISCUSSION

Far from being an anomaly, informality is the norm across cities in the southern African region. As formal urban planning and development processes seek to promote resilience, it is critical that they grapple with the complex ways in which resilience dynamics are shaped by informal land use, informal economic activities, informal institutional arrangements, and even designations of people as "informal." This requires attention to how the structure of informality varies between countries in the region. For example, urban resilience work is likely to entail different considerations in Zimbabwe, where the economy is mostly informal but a minority of the population lives in informal settlements, and Lesotho, where these parameters are reversed (Fig. 1). Our cases illustrate how that informality can be associated with exacerbated exposure to risk for both people's homes and incomes through a lack of access to publicly regulated water and sanitation infrastructure, land tenure insecurity, and poor housing options. Yet, the absence of formal actors and institutions also gives rise to innovation, with both local and systemic social-ecological implications. Affluent households find ways to source their own water during drought, vendors trade illegally to secure livelihoods, and people settle precariously on riverbanks as they have little or no access to the formal system of land allocation. These strategies and innovations (particularly those that substitute for actions normally assumed by public authorities, or that generate positive systemic externalities) are embedded in social-political relations. They often entail inequities in relation to burdens and benefits, as well as raise questions of responsibility and accountability. We discuss these issues below.

Making informality visible and confronting power

Informality is everywhere: it is part and parcel of economic transactions, natural resource use, and is often the means through which diverse populations—affluent and poor, in gated communities or unregulated settlements—innovate to meet critical needs (Fig. 2). However, official statistics and measures rarely capture this ubiquity, often leaving informality invisible until a crisis strikes. At the heart of this paradox is the critical role of recognitional justice (Bulkeley et al. 2014). Formal urban governance institutions largely have the power to determine what is included or excluded from urban services, what rights (and whose) are acknowledged and formally upheld, and which places and populations have claims to urban resources (Ziervogel et al. 2017).

Fig. 2. The informal is not "somewhere else" in a city – it is everywhere and underpins most socioeconomic activities. By graying out the formal parts of the urban landscape, the lower panel highlights informal activities, several of which (e.g., B, C, D) are not typically counted as "informal" in official metrics. Four specific examples are discussed in the main text: (A) sand mining, (B) disaster relief during flooding, (C) governance and decision-making, and (D) unregistered boreholes.

Which informal activities, or actors involved in such activities, are "recognized" in formal governance or formal economic activities, are political and contested choices (Alfaro d'Alençon et al. 2018). In Lilongwe, public sector and formally recognized real estate actors are implicated in informal land transactions and construction supply chains (Fig. 2A) that exacerbate systemic exposure to flooding (Makuwira 2022). This undermines both locally experienced and city-scale resilience, echoing similar dynamics elsewhere (Weinstein 2008, Lambert 2020, Tellman et al. 2021). In Cape Town, affluent residents, in formally zoned areas with access to municipal water, secure additional water informally for non-potable use via unregistered boreholes (Fig. 2D), an activity that becomes normatively accepted and perceived as a formal right of property ownership. In contrast, informal food traders in Lilongwe and backyard dwellers in Cape Town struggle to get their interests legitimized and recognized. Thus, what is made visible, and what remains hidden or considered illegitimate, depends on how actors in power view the desirability of that particular form of informality.

Labeling activities and, most important, people, as "informal" is ultimately a process of othering and "moral exclusion" (Opotow 1990). When city authorities formally recognize informal activities, there is an assumption that they will assume some responsibility for any emergent externalities associated with these activities that undermine resilience. Denying such recognition thus buries both positive and negative externalities resulting from these activities from "formal" responsibility and purview (Eakin et al. 2016).

The informal can enhance or undermine resilience

The connections between local- and city-level resilience are complex, and made even more so by relations of informality. In Cape Town, the informal extraction of groundwater by affluent residents enhances their immediate and longer-term resilience during the drought, but the broader impacts on the city's revenue is negative, and there is considerable uncertainty about the longer-term sustainability of underground aquifers and the roles these play in regional ecosystems and water security (WWF 2020). Similarly, Lilongwe's informal waste management and sand mining may provide immediate economic benefits for some residents, but over time they increase the risk of disease and flooding for the city and communities further downstream. Viewing the informal and formal as an intertwined whole, interacting across spatial and temporal scales, forces reflexivity around setting system boundaries for resilience planning (Harris et al. 2018).

Systems are in constant flux, punctuated by crises when functional boundaries of systems can become visible and provide opportunities for learning and innovation in governance (Eakin et al. 2023). The adaptive cycle heuristic describes social-ecological systems as they move through phases of growth, accumulation, restructuring (often after some kind of shock or release), and renewal or transformation (Gunderson and Holling 2002; see also Rocha et al. 2022). Informal activities can play critical roles in urban systems' adaptive cycle, especially as innovators and pioneers who can use newly available niches or provide services lacking when formal responses are absent. Learning is fundamental during such times (Eakin et al. 2023, Hamann et al. 2023), as in the case of Cape Town's post-drought re-think of water supplies and new strategy to build system resilience against future droughts.

Nevertheless, while system dynamics (and particularly crises) can provide opportunity for recognition, innovation, and learning, this does not always happen. In resilience theory, taking advantage of "windows of opportunity" often requires adaptive or transformative capacity, including resources and bridging organizations that facilitate networking and collaboration (Olsson et al. 2006, Wolfram et al. 2019). Despite repeated crises in Lilongwe, the governance system's response has barely changed. Instead, the externally supported informal institutions provide an invisible backbone of adaptive capacity during crises, allowing the formal system to remain in the conservation phase with little incentive to reform or innovate (Dovey 2012).

Innovations in governance may hinge on organizations that can bridge informal/formal divides

While the formal/informal distinction is mutable, porous, and politically created, once it has been applied and even institutionalized, governing across it can be very difficult.

Approaches to governance range from expanding the regulatory reach of the state ("regularization"), legal recognition and incorporation of informality ("formalization"), or embracing the innovation emerging in informality through recognition, support, and policy reform (Brown et al. 2014).

Bridging organizations and intermediary actors connect distinct spheres of governance or influence (Crona and Parker 2012, de Kraker 2017). They can help coordinate action between actors who lack resources, capacity, mandates, or even interest in collaborating (Rathwell and Peterson 2012). Bridging organizations that are formally constituted (typically registered non-profits and civil society groups) can have the legal and institutional standing to command the attention of government and the public through media and advocacy. Bridging organizations are key players in building system resilience through facilitating learning and knowledge sharing across stakeholders with distinct values and norms, and most effectively influence governance when they can build relations of trust, navigate asymmetrical power relations, and create depoliticized spaces for negotiation (Crona and Parker 2012). This work can enable cities to experiment with service co-production across informal-formal divides in an effort to solve urban problems (Fig. 2C; Brown et al. 2014). By having a foot in both worlds, such organizations, and sometimes single individuals with leadership capacity, may form bridges that enable a merging of top-down/bottom-up binaries, and thus forming more equitable approaches to urban resilience planning and policy (Sarmiento and Tilly 2018, Ziervogel 2019, Enqvist et al. 2020).

Such brokerage can be direct, facilitating explicit partnerships and alliances where people representing voices from informal activities and spaces are brought into formal decision-making processes (Nagendra et al. 2018, Ziervogel 2019, Ziervogel et al. 2021). One example is the ICLEI–brokered, jointly developed river restoration plan where informal vendors and the Lilongwe City Council partnered to minimize market waste disposal in the river. In another case, an NGO in Lusaka (Waste Recyclers Association of Zambia) advocated for informal waste pickers' interests, safety needs, and livelihood priorities in negotiations about the city's waste management policy (Chileshe and Moonga 2017). By supporting the political agency of informal groups and activists, bridging organizations can help ensure more equitable negotiations of roles and responsibilities across informal and formal spaces. This creates opportunities for not only recognition and procedural justice but also institutional innovation (Brown et al. 2014, Sarmiento and Tilly 2018).

Beyond having legitimacy, trust, and respect among actors in both formal and informal systems, bridging organizations need to have influence in decision-making at multiple organizational levels (province, city, neighborhood), and skills to convene disparate groups and broker differences in knowledge, values, and needs (Crona and Parker 2012). Without attention to recognitional justice, bridging organizations may inadvertently leave formal actors unaccountable to informal systems. This may be occurring in Lilongwe, where external support for informal disaster response (Fig. 2B) may enable the continuation of inequitable policies that exclude some people and areas from public services and risk management. Bridging organizations are also subject to the power and perceived legitimacy of different stakeholders, and the expressed urgency of their needs (Crona and Parker 2012). A city struggling to meet urgent public safety needs along an urban river may therefore have greater sway over residents of a riparian informal settlement advocating for greater economic opportunities. Thus, although clearly there is a role for these intermediaries, the specific attributes that make them effective bridging organizations between informal and formal spheres require more attention (de Kraker 2017).

Enhancing collective capacities to address equity in urban system resilience

Acknowledging the informal as part and parcel of urban fabric can create opportunities for governance innovations to strengthen urban resilience. Yet, for these to emerge and be sustained, both formal and informal sectors and actors need financial, technical, and political capacity (Dodman and Satterthwaite 2008). Engaging with both formal and informal activities and spaces in the multi-level and complex temporal dynamics of urban resilience is an enormous challenge. It involves anticipating the impacts and social-spatial distribution of shocks and stressors, understanding and accounting for complex system feedback and thresholds of change, and gaining insight into how local level actions embedded within larger urban economic and demographic processes can affect resilience outcomes. Such "resilience thinking" requires diverse knowledge and often unconventional ways of approaching problems.

Most cities in the Global South, and many in the Global North, are financially constrained (Serageldin et al. 2008, Smoke 2017) and face institutional and jurisdictional limits to what they can or cannot do in relation to informal activities, settlements, or labor (Lehmann et al. 2015, Hernández Aguilar et al. 2021). Cities may be prohibited from offering services to informal settlements, as mandates to do so reside with regional or national agencies, or because technical capacities are lacking (Tellman et al. 2021). Across the formal-informal continuum, actors often also lack the experience and expertise in equitable social engagement, undermining well-intentioned initiatives (Wolfram et al. 2019).

These capacity gaps can be filled by advocacy organizations that help design procedures that foster recognition of the rights of different actors in the urban space, and that bring attention to inequities in resilience work. Academic and non-profit groups can assist urban administrators in designing modes of engagement that are conducive to learning in both informal and formal social and spatial contexts. For example, experimental approaches such as "Learning Journeys" or "Transformation Laboratories" and "sense-making" have demonstrated that the process of bridging informal and formal divides is as important as the outcome (Enqvist et al. 2020, Pereira et al. 2022). In Lilongwe, an intermediary organization, ICLEI, encouraged the city to work with an informal site for a river restoration project (as part of SwedBio's Urban Natural Assets Programme) as a means of shifting officials' perceptions about how best to work with informality. Through a series of trust-building engagements the city, the organization, and local communities collectively developed ideas on ways to manage waste at the site and restore the river system (ICLEI 2020). In another example, the resilience strategy of Accra, Ghana, intentionally and explicitly integrated the perspective of residents in informal settlements and those involved in the informal economy in its planning process. The result is a strategy that "embraces informality in its urban system" as one of the three strategic pillars (Accra Metropolitan Assembly 2019), where bridging organizations like the Slum Union of Ghana, Slum Dwellers International and the People's Dialogue on Human Settlements have been instrumental in achieving the recognition and procedural justice expressed in the strategy (Cobbinah and Finn 2023). Nevertheless, implementation of plans to foreground informality in enhancing resilience are partially impeded because authorities "lack the confidence to engage, integrate, and build on informal grassroots initiatives and networks to deliver transformative climate adaptation" (Cobbinah and Finn 2023:376).

CONCLUSION

Cities are inherently complex and dynamic. Informality is an inextricable part of the urban fabric, particularly in the Global South. As cities turn to resilience as part of their policy efforts in pursuing more sustainable and equitable future development, informality in all its complexity needs to take center stage. Informality cannot be reduced to the spatial margins of urban administration, or the social exceptions to the rule of law and order—it is interdependent with formal policy processes, embedded within formal spaces, and an emergent property of the decisions made by "formal" entities and state actors. Planning for resilience for urban systems must involve grappling with this complexity if resilience is to reflect equity and social justice. How are the boundaries of urban systems defined and who or what is made invisible in such definitions? Whose voices are excluded or "mis-recognized" and why? What activities, undertaken by what actors, are villainized or sanctioned? And who bears the burdens and benefits of innovation, problem solving, and managing the inevitable risks of urban development?

Our analysis illustrates how informality of place, activity, and in rules and norms can simultaneously serve to undermine resilience (pushing social-ecological systems toward thresholds of undesirable change) as well as reinforce resilience by enabling capacities to manage disturbance, respond proactively to opportunity, and innovate to ensure viability. That said, our analysis is based on a literature synthesis and the collective experience and meaning-making of a group of researchers and practitioners familiar with the southern African context. It is not a systematic review or quantitative case study analysis, meaning that this paper's arguments should be seen as a call for more research on informality's role in urban resilience rather than a conclusive account of it. How informality's relation to resilience is interpreted will depend on spatial and temporal scale, relations of power, and how the benefits and burdens of actions to manage risk and change are distributed. Shining a spotlight on the diverse informal spaces and processes that constitute and contribute to urban dynamics may not always be desired or welcomed. Nevertheless, ignoring the ubiquity of informality and the complex roles and implications it has in urban systems is also problematic. Trusted intermediaries (boundary organizations that can legitimately bridge the fuzzy but undeniable formal/informal divide) can help navigate the complex politics of recognition, inclusion and legitimization that are critical in pursuing more equitable urban resilience.

Author Contributions:

This manuscript is a result of a collective, collaborative effort; we therefore list the authors alphabetically. All authors contributed to the conceptualization and writing of the paper. HE, JE, and MH contributed mainly to the theoretical background and development of the figures; NM led the policy review; EvW, JS, MS, JE, and GZ compiled the case studies; and all authors contributed to the discussion.

Acknowledgments:

This paper was developed as part of the Southern African Resilience Academy (SARA), an initiative of the Global Resilience Partnership, with support from the South African Research Chairs Initiative (SARChI) of the Department of Science and Technology and National Research Foundation of South Africa (grant 98766). Swedbio is gratefully acknowledged for funding ICLEI's work in Lilongwe through the Urban Natural Assets Programme. The authors wish to thank their fellow SARA participants for feedback on early drafts of the paper, Mqondisi Ngwenya for research on existing urban resilience strategies, Slum Dwellers International / CCODE for case study material, as well as the anonymous reviewers for their constructive comments. HE acknowledges the support of the United States Fulbright Scholars Program 2021-22. JE acknowledges the support of Vetenskapsrådet grant no. 2019-00508. Graphics created by KANDS Collective (hello@kandscollective.com).

Data Availability:

As a synthesis article, this paper uses only second-hand data. All sources are stated in the reference list.

LITERATURE CITED

Adams, E. A. 2017. Thirsty slums in African cities: household water insecurity in urban informal settlements of Lilongwe, Malawi. International Journal of Water Resources Development 34(6):869-887. https://doi.org/10.1080/07900627.2017.1322941

Accra Metropolitan Assembly. 2019. Accra resilient strategy. AMA, Accra, Ghana. https://ama.gov.gh/documents/Accra-Resilience-Strategy.pdf

African Cities Research Consortium (ACRC) 2021. Lilongwe City scoping study. African Cities Research Consortium.

Ahlers, R., F. Cleaver, M. Rusca, and K. Schwartz. 2014. Informal space in the urban waterscape: disaggregation and co-production of water services. Water Alternatives 7(1):1-14.

Akinola, S. R. 2016. Urbanization, collective action and coping strategies in informal areas of African Cities: a polycentric environmental planning perspective. Pages 5-24 in S. Attia, S. Shabka, Z. Shafik, and A. Ibrahim, editors. Dynamics and resilience of informal areas. Springer, Cham, Switzerland. https://doi.org/10.1007/978-3-319-29948-8_1

Alda-Vidal, C., M. Kooy, and M. Rusca. 2017. Mapping operation and maintenance: an everyday urbanism analysis of inequalities within piped water supply in Lilongwe, Malawi. Urban Geography 39(1):104-121. https://doi.org/10.1080/02723638.2017.1292664

Alfaro d'Alençon, P., H. Smith, E. Álvarez de Andrés, C. Cabrera, J. Fokdal, M. Lombard, A. Mazzolini, E. Michelutti, L. Moretto, and A. Spire. 2018. Interrogating informality: conceptualisations, practices and policies in the light of the New Urban Agenda. Habitat International 75:59-66. https://doi.org/10.1016/j.habitatint.2018.04.007

Auerbach, A. M., A. LeBas, A. E. Post, and R. Weitz-Shapiro. 2018. State, society, and informality in cities of the Global South. Studies in Comparative International Development 53:261-280. https://doi.org/10.1007/s12116-018-9269-y

Banks, N., M. Lombard, and D. Mitlin. 2020. Urban informality as a site of critical analysis. Journal of Development Studies 56(2):223-238. https://doi.org/10.1080/00220388.2019.1577384

Battersby, J., M. Marshak, and N. Mngqibisa. 2016. Mapping the informal food economy in Cape Town, South Africa. HCP Working Papers. Social Sciences and Humanities Research Council of Canada.

Brown, D., G. McGranahan, and D. Dodman. 2014. Urban informality and building a more inclusive, resilient and green economy. IIED working paper. IIED, London, UK.

Budlender, D., and S. Gondwe. 2019. Budgeting and the informal economy in Lilongwe, Malawi. Budget Brief No. 8. WIEGO, Manchester, UK.

Bulkeley, H., G. A. S. Edwards, and S. Fuller. 2014. Contesting climate justice in the city: examining politics and practice in urban climate change experiments. Global Environmental Change 25:31-40. https://doi.org/10.1016/j.gloenvcha.2014.01.009

Chapman, R. A., and N. Sasman. 2012. Cities and climate change: urban vulnerability and resilience in Southern Africa. Regional Climate Change Programme Southern Africa (RCCP)/UK Department for International Development, Cape Town, South Africa.

Charmes, J. 2019. Dimensions of resilience in developing countries: informality, solidarities and carework. Springer, Paris, France. https://doi.org/10.1007/978-3-030-04076-5

Chileshe, B., and M. S. Moonga. 2017. Alternatives for dumpsite scavenging: the case of waste pickers at Lusaka's Chunga landfill. International Journal of Humanities, Social Sciences and Education 4(6):40-51. https://doi.org/10.20431/2349-0381.0406006

Chigwenya, A., and P. Wadzanai. 2020. The intersectionality between right to the city, informality, and waste management in Masvingo City, Zimbabwe. Journal of Urban Systems and Innovations for Resilience in Zimbabwe 2(2):26-52.

City of Cape Town 2019. Our Shared Water Future: Cape Town's Water Strategy. City of Cape Town, South Africa.

Cobbinah, P. B., and B. M. Finn. 2022. Planning and climate change in African cities: informal urbanization and 'just' urban transformations. Journal of Planning Literature 38(3):361-379. https://doi.org/10.1177/08854122221128762

Cote, M., and A. J. Nightingale. 2011. Resilience thinking meets social theory: situating social change in socio-ecological systems (SES) research. Progress in Human Geography 36(4):475-489. https://doi.org/10.1177/0309132511425708

Crona, B. I., and J. N. Parker. 2012. Learning in support of governance: theories, methods, and a framework to assess how bridging organizations contribute to adaptive resource governance. Ecology and Society 17(1):32. https://doi.org/10.5751/ES-04534-170132

Dade, M. C., A. S. Downing, K. Benessaiah, M. Falardeau, M. Lin, J. T. Rieb, and J. C. Rocha. 2022. Inequalities in the adaptive cycle: reorganizing after disasters in an unequal world. Ecology and Society 27(4):10. https://doi.org/10.5751/ES-13456-270410

Daniels, S. 2010. Making do: innovation in Kenya's informal economy. Analogue Digital, New York, New York, USA.

Darvas, P., S. Gao, Y. Shen, and B. Bawany. 2017. Sharing higher education's promise beyond the few in sub-Saharan Africa. World Bank Publications.

de Kraker, J. 2017. Social learning for resilience in social-ecological systems. Current Opinion in Environmental Sustainability 28:100-107. https://doi.org/10.1016/j.cosust.2017.09.002

Dodman, D., and D. Satterthwaite. 2008. Institutional capacity, climate change adaptation and the urban poor. IDS Bulletin 39:67-74. https://doi.org/10.1111/j.1759-5436.2008.tb00478.x

Dovey, K. 2012. Informal urbanism and complex adaptive assemblage. International Development Planning Review 34(4):349-368. https://doi.org/10.3828/idpr.2012.23

Eakin, H., R. Hamann, G. Ziervogel, and C. Shearing. 2023. Emergent governance responses to shocks to critical provisioning systems. npj Urban Sustainability 3:42. https://doi.org/10.1038/s42949-023-00123-y

Eakin, H., A. M. Lerner, D. Manuel-Navarrete, B. Hernández Aguilar, A. Martínez-Canedo, B. Tellman, L. Charli-Joseph, R. Fernández Álvarez, and L. Bojórquez-Tapia. 2016. Adapting to risk and perpetuating poverty: household's strategies for managing flood risk and water scarcity in Mexico City. Environmental Science and Policy 66:324-333. https://doi.org/10.1016/j.envsci.2016.06.006

Eakin, H. C., J. Parajuli, B. Hernández Aguilar, and Y. Yogya. 2022. Attending to the social-political dimensions of urban flooding in decision-support research: a synthesis of contemporary empirical cases. WIREs Climate Change 13(1):e743. https://doi.org/10.1002/wcc.743

Elmqvist, T., E. Andersson, N. Frantzeskaki, T. McPhearson, P. Olsson, O. Gaffney, K. Takeuchi, and C. Folke. 2019. Sustainability and resilience for transformation in the urban century. Nature Sustainability 2:267-273. https://doi.org/10.1038/s41893-019-0250-1

Enqvist, J. P., and G. Ziervogel. 2019. Water governance and justice in Cape Town: an overview. WIREs Water 6(4):e1354. https://doi.org/10.1002/wat2.1354

Enqvist, J., G. Ziervogel, L. Metelerkamp, J. van Breda, N. Dondi, T. Lusithi, A. Mdunyelwa, Z. Mgwigwi, M. Mhlabisi, S. Myeza, et al. 2020. Informality and water justice: community perspectives on water issues in Cape Town's low-income neighbourhoods. International Journal of Water Resources Development 38 (1):108-129. https://doi.org/10.1080/07900627.2020.1841605

Fandamu, M. N., and H. Fandamu. 2018. The effects of sand mining on poverty status in Malawi: a case of Lilongwe urban. International Journal of Social Sciences, Humanities and Education 2:87-104.

Finn, B. M., and P. B. Cobbinah. 2022. African urbanisation at the confluence of informality and climate change. Urban Studies 60(3):405-424. https://doi.org/10.1177/00420980221098946

Fitzgibbons, J., and C. L. Mitchell. 2019. Just urban futures? Exploring equity in "100 Resilient Cities." World Development 122:648-659. https://doi.org/10.1016/j.worlddev.2019.06.021

Frantzeskaki, N., S. Buchel, C. Spork, K. Ludwig, and M. T. J. Kok. 2019. The multiple roles of ICLEI: intermediating to innovate urban biodiversity governance. Ecological Economics 164:106350. https://doi.org/10.1016/j.ecolecon.2019.06.005

Garcia, A., N. Gonda, E. Atkins, N. J. Godden, K. P. Henrique, M. Parsons, P. Tschakert, and G. Ziervogel. 2022. Power in resilience and resilience's power in climate change scholarship. WIREs Climate Change 13(3):e762. https://doi.org/10.1002/wcc.762

Global Center on Adaptation. 2022. Stories of resilience: lessons from local practice. Rotterdam, the Netherlands.

Global Informality Project. 2024. What is informality? UCL School of Slavonic and East European Studies. https://www.in-formality.com/wiki/index.php?title=What_is_informality%3F

Gunderson, L. H., and C. S. Holling. 2002. Panarchy: understanding transformation in human and natural systems. Island Press, Washington, D.C., USA.

Hamann, R., J. Soderbergh, A. Surmeier, C. Fyvie, T. Ramarumo, M. Rapson, A. Newell, and N. Sitas. 2023. Building solidarity in crisis: why and how emergent groups organize cross-boundary collective action. Academy of Management Proceedings 2023:17014. https://doi.org/10.5465/AMPROC.2023.17014abstract

Harris, L. M., E. K. Chu, and G. Ziervogel. 2018. Negotiated resilience. Resilience 6(3):196-214. https://doi.org/10.1080/21693293.2017.1353196

Hernández Aguilar, B., A. M. Lerner, D. Manuel-Navarrete, and J. M. Siqueiros-García. 2021. Persisting narratives undermine potential water scarcity solutions for informal areas of Mexico City: the case of two settlements in Xochimilco. Water International 46(6):919-937. https://doi.org/10.1080/02508060.2021.1923179

ICLEI. 2020. Working with informality: promoting collaborative efforts to service informal settlements with nature-based solutions. Urban Natural Assets for Africa Programme.

Intergovernmental Panel on Climate Change (IPCC). 2023. Africa. In: Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Pages 1285-1456. Cambridge University Press, Cambridge, UK.

International Labor Organization (ILO). 2023. Statistics on the informal economy. International Labor Organization. https://ilostat.ilo.org/topics/informality/

Jack, K., M. Visser, E. Robertson, and K. Berger. 2019. Day zero-understanding responses to the Cape Town water crisis. City of Cape Town, South Africa.

Johannessen, Å. and C. Wamsler. 2017. What does resilience mean for urban water services? Ecology and Society 22(1):1. https://doi.org/10.5751/ES-08870-220101

Kita, S. M. 2017. "Government doesn't have the muscle": state, NGOs, local politics, and disaster risk governance in Malawi. Risk, Hazards and Crisis in Public Policy 8(3):244-267. https://doi.org/10.1002/rhc3.12118

Kooy, M. 2014. Developing informality: the production of Jakarta's urban waterscape. Water Alternatives 7:35-53.

Kumar, H., and S. Bhaduri. 2014. Jugaad to grassroot innovations: understanding the landscape of the informal sector innovations in India. African Journal of Science, Technology, Innovation and Development 6(1):13-22. https://doi.org/10.1080/20421338.2014.895481

Lambert, R. 2020. Land trafficking and the fertile spaces of legality. International Journal of Urban and Regional Research 45(1):21-38. https://doi.org/10.1111/1468-2427.12975

Leach, M., B. Reyers, X. Bai, E. S. Brondizio, C. Cook, S. Díaz, G. Espindola, M. Scobie, M. Stafford-Smith, and S. M. Subramanian. 2018. Equity and sustainability in the Anthropocene: a social-ecological systems perspective on their intertwined futures. Global Sustainability 1:e13. https://doi.org/10.1017/sus.2018.12

Lehmann, P., M. Brenck, O. Gebhardt, S. Schaller, and E. Süßbaue. 2015. Barriers and opportunities for urban adaptation planning: analytical framework and evidence from cities in Latin America and Germany. Mitigation and Adaptation Strategies for Global Change 20:75-97. https://doi.org/10.1007/s11027-013-9480-0

Lilongwe Water Board. 2021. Sanitation and drainage improvement strategy and institutional framework for Lilongwe City (draft). Lilongwe, Malawi.

Losch, B. 2016. Structural transformation to boost youth labour demand in sub-Saharan Africa: the role of agriculture, rural areas and territorial development. International Labour Organization, Geneva, Switzerland.

Mahlanza, L., G. Ziervogel, and D. Scott. 2016. Water, rights and poverty: an environmental justice approach to analysing water management devices in Cape Town. Urban Forum 27:363-382. https://doi.org/10.1007/s12132-016-9296-6

Makuwira, J. 2022. Marginal voices, resilient acts: urban marginality and responses to climate-related events in Lilongwe City informal settlements. Urban Forum 33:51-64. https://doi.org/10.1007/s12132-022-09460-0

Malebana, D. S. 2021. Sand mining, land degradation, and rehabilitation in rural areas of South Africa: a case of Mentz village, Limpopo Province. Development Studies in Planning and Management, University of Limpopo, South Africa.

Matin, N., J. Forrester, and J. Ensor. 2018. What is equitable resilience? World Development 109:197-205. https://doi.org/10.1016/j.worlddev.2018.04.020

McClymont, K., and A. Sheppard. 2020. Credibility without legitimacy? Informal development in the highly regulated context of the United Kingdom. Cities 97:102520. https://doi.org/10.1016/j.cities.2019.102520

McFarlane, C. 2012. Rethinking informality: politics, crisis, and the city. Planning Theory and Practice 13(1):89-108. https://doi.org/10.1080/14649357.2012.649951

McPhearson, T., C. M. Raymond, N. Gulsrud, C. Albert, N. Coles, N. Fagerholm, M. Nagatsu, A. S. Olafsson, N. Soininen, and K. Vierikko. 2021. Radical changes are needed for transformations to a good Anthropocene. npj Urban Sustainability 1:5. https://doi.org/10.1038/s42949-021-00017-x

Meerow, S., and J. P. Newell. 2016. Urban resilience for whom, what, when, where, and why? Urban Geography 40(3):309-329. https://doi.org/10.1080/02723638.2016.1206395

Meerow, S., J. P. Newell, and M. Stults. 2016. Defining urban resilience: a review. Landscape and Urban Planning 147:38-49. https://doi.org/10.1016/j.landurbplan.2015.11.011

Meerow, S., P. Pajouhesh, and T. R. Miller. 2019. Social equity in urban resilience planning. Local Environment 24(9):793-808. https://doi.org/10.1080/13549839.2019.1645103

Mitlin, D., and A. Walnycki.2019. Informality as experimentation: water utilities' strategies for cost recovery and their consequences for universal access. Journal of Development Studies 56 (2):259-277. https://doi.org/10.1080/00220388.2019.1577383

Nagendra, H., X. Bai, E. S. Brondizio, and S. Lwasa. 2018. The urban south and the predicament of global sustainability. Nature Sustainability 1:341-349. https://doi.org/10.1038/s41893-018-0101-5

Ohnsorge, F., and S. Yu, editors. 2022. The long shadow of informality: challenges and policies. World Bank Publications. https://doi.org/10.1596/35782

Olsson, P., L. H. Gunderson, S. R. Carpenter, P. Ryan, L. Lebel, C. Folke, and C. S. Holling. 2006. Shooting the rapids: navigating transitions to adaptive governance of social-ecological systems. Ecology and Society 11(1):18. https://doi.org/10.5751/ES-01595-110118

Opotow, S. 1990. Moral exclusion and injustice: an introduction. Journal of Social Issues 46(1):1-20. https://doi.org/10.1111/j.1540-4560.1990.tb00268.x

Parthasarathy, D. 2015. Informality, resilience, and the political implications of disaster governance. Pacific Affairs 88 (3):551-575. https://doi.org/10.5509/2015883551

Pascual, U., J. Phelps, E. Garmendia, K. Brown, E. Corbera, A. Martin, E. Gomez-Baggethun, and R. Muradian. 2014. Social equity matters in payments for ecosystem services. BioScience 64 (11):1027-1036. https://doi.org/10.1093/biosci/biu146

Pereira, L. M., S. B. Kushitor, C. Cramer, S. Drimie, M. Isaacs, R. Malgas, E. Phiri, C. Tembo, and J. Willis. 2022. Leveraging the potential of wild food for healthy, sustainable, and equitable local food systems: learning from a transformation lab in the Western Cape region. Sustainability Science. https://doi.org/10.1007/s11625-022-01182-3

Peters, L. E. R., A. Clark-Ginsberg, B. McCaul, G. Cáceres, A. L. Nuñez, J. Balagna, A. López, S. S. Patel, R. B. Patel, and J. Van Den Hoek. 2022. Informality, violence, and disaster risks: coproducing inclusive early warning and response systems in urban informal settlements in Honduras. Frontiers in Climate 4:937244. https://doi.org/10.3389/fclim.2022.937244

Ramkumar, M., K. Kumaraswamy, R. A. James, M. Suresh, T. Sugantha, L. Jayaraj, A. Mathiyalagan, M. Saraswathi, and J. Shyamala. 2015. Sand mining, channel bar dynamics and sediment textural properties of the Kaveri River, South India: implications on flooding hazard and sustainability of the natural fluvial system. Pages 283-318 in M. Ramkumar, K. Kumaraswamy, and R. Mohanraj, editors. Environmental management of river basin ecosystems. Springer International Publishing, Cham, Switzerland. https://doi.org/10.1007/978-3-319-13425-3_14

Rathwell, K. J., and G. D. Peterson. 2012. Connecting social networks with ecosystem services for watershed governance: a social-ecological network perspective highlights the critical role of bridging organizations. Ecology and Society 17(2):24. https://doi.org/10.5751/ES-04810-170224

Revi, A., D. Roberts, I. Klaus, A. Bazaz, J. Krishnaswamy, C. Singh, A. Eichel, P. P. Kodira, S. Seth, I. Adelekan, et al. 2022. The summary for urban policymakers of the IPCC's Sixth Assessment Report. Indian Institute for Human Settlements. https://doi.org/10.24943/SUPSV511.2022

Robins, S. 2019. 'Day Zero', hydraulic citizenship and the defence of the commons in Cape Town: a case study of the politics of water and its infrastructures (2017-2018). Journal of Southern African Studies 45(1):5-29. https://doi.org/10.1080/03057070.2019.1552424

Rocha, J. C., L. B. Luvuno, J. T. Rieb, E. T. H. Crockett, K. Malmborg, M. Schoon, and G. D. Peterson. 2022. Panarchy: ripples of a boundary concept. Ecology and Society 27(3):21. https://doi.org/10.5751/ES-13374-270321

Roy, A. 2005. Urban informality: toward an epistemology of planning. Journal of the American Planning Association 71 (2):147-158. https://doi.org/10.1080/01944360508976689

Roy, A. 2009. Why India cannot plan its cities: informality, insurgence and the idiom of urbanization. Planning Theory 8 (1):76-87. https://doi.org/10.1177/1473095208099299

Šakić Trogrlić, R., G. B. Wright, A. J. Adeloye, M. J. Duncan, and F. Mwale. 2017. Taking stock of community-based flood risk management in Malawi: different stakeholders, different perspectives. Environmental Hazards 17(2):107-127. https://doi.org/10.1080/17477891.2017.1381582

Sandoval, V., and J. P. Sarmiento. 2020. A neglected issue: informal settlements, urban development, and disaster risk reduction in Latin America and the Caribbean. Disaster Prevention and Management 29(5):731-745. https://doi.org/10.1108/DPM-04-2020-0115

Sarmiento, H., and C. Tilly. 2018. Governance lessons from urban informality. Politics and Governance 6(1):199-202. https://doi.org/10.17645/pag.v6i1.1169

Schlosberg, D. 2004. Reconceiving environmental justice: global movements and political theories. Environmental Politics 1 (3):517-540. https://doi.org/10.1080/0964401042000229025

Schlosberg, D. 2007. Defining environmental justice: theories, movements, and nature. Oxford University Press, Oxford, UK. https://doi.org/10.1093/acprof:oso/9780199286294.001.0001

Serageldin, M., D. Jones, F. Vigier, and E. Solloso. 2008. Municipal financing and urban development. Human Settlements Global Dialogue Series, No. 3. Nairobi, Kenya: Center for Urban Development Studies, Harvard Design School. UN-HABITAT, Nairobi, Kenya.

Sheppard, E., T. Sparks, and H. Leitner. 2020. World class aspirations, urban informality, and poverty politics: a north-south comparison. Antipode 52(2):393-407. https://doi.org/10.1111/anti.12601

Simatele, D. M., S. Dlamini, and N. S. Kubanza. 2017. From informality to formality: perspectives on the challenges of integrating solid waste management into the urban development and planning policy in Johannesburg, South Africa. Habitat International 63:122-130. https://doi.org/10.1016/j.habitatint.2017.03.018

Simpson, N. P., C. D. Shearing, and B. Dupont. 2019a. Climate gating: a case study of emerging responses to Anthropocene Risks. Climate Risk Management 26:100196. https://doi.org/10.1016/j.crm.2019.100196

Simpson, N. P., C. D. Shearing, and B. Dupont. 2020. 'Partial functional redundancy': an expression of household level resilience in response to climate risk. Climate Risk Management 28:100216. https://doi.org/10.1016/j.crm.2020.100216

Simpson, N. P., K. J. Simpson, C. D. Shearing, and L. R. Cirolia. 2019b. Municipal finance and resilience lessons for urban infrastructure management: a case study from the Cape Town drought. International Journal of Urban Sustainable Development 11(3):257-276. https://doi.org/10.1080/19463138.2019.1642203

Smoke, P. 2017. Strengthening an overlooked foundation of urban planning. In G. Bhan, S. Srinivas, and V. Watson, editors. The Routledge companion to planning in the Global South. Routledge, New York, USA.

Sultana, F. 2021. Climate change, COVID-19, and the co-production of injustices: a feminist reading of overlapping crises. Social and Cultural Geography 22(4):447-460. https://doi.org/10.1080/14649365.2021.1910994

Taylor, A., and C. Peter. 2014. Strengthening climate resilience in African cities: a framework for working with informality. Climate Development Knowledge Network, African Center for Cities, Cape Town, South Africa.

Tellman, B., H. Eakin, M. A. Janssen, F. de Alba, and B. L. Turner II. 2021. The role of institutional entrepreneurs and informal land transactions in Mexico City's urban expansion. World Development 140:105374. https://doi.org/10.1016/j.worlddev.2020.105374

United Nations (UN). 2018. The world's cities in 2018–data booklet (ST/ESA/SER.A/417). United Nations Department of Economic and Social Affairs, Population Division, New York, New York, USA.

UN-Habitat. 2011. Malawi- Lilongwe Urban Profile. UN-Habitat.

UN-Habitat. 2015. Slum Almanac 2015/2016. UN-Habitat, Nairobi, Kenya.

Weinstein, L. 2008. Mumbai's development mafias: globalization, organized crime and land development. International Journal of Urban and Regional Research 32(1):22-39. https://doi.org/10.1111/j.1468-2427.2008.00766.x

Whaley, L. 2018. The critical institutional analysis and development (CIAD) framework. International Journal of the Commons 12(2):137-161. https://doi.org/10.18352/ijc.848

Wolfram, M., S. Borgström, and M. Farrelly. 2019. Urban transformative capacity: from concept to practice. Ambio 48:437-448. https://doi.org/10.1007/s13280-019-01169-y

Wood, G. 2003. Staying secure, staying poor: the "Faustian bargain." World Development 31(3):455-471. https://doi.org/10.1016/S0305-750X(02)00213-9

World Bank. 2019. Managing risks for a safer built environment in Malawi. Building Regulatory Capacity Assessment. World Bank, Washington, D.C., USA. https://doi.org/10.1596/32761

World Bank. 2023. Population living in slums. https://data.worldbank.org/indicator/EN.POP.SLUM.UR.ZS?name_desc=false

World Population Review. 2023. Lilongwe - population. https://worldpopulationreview.com/world-cities/lilongwe-population

WWF. 2020. Cape Town groundwater fact sheet. World Wildlife Fund, Cape Town, South Africa.

Ziervogel, G. 2019. Unpacking the Cape Town drought: lessons learned. Report for Cities Support Programme. African Centre for Cities, Cape Town, South Africa.

Ziervogel, G., J. Enqvist, L. Metelerkamp, and J. Van Breda. 2021. Supporting transformative climate adaptation: community-level capacity building and knowledge co-creation in South Africa. Climate Policy 22(5):607-622. https://doi.org/10.1080/14693062.2020.1863180

Ziervogel, G., M. Pelling, A. Cartwright, E. Chu, T. Deshpande, L. Harris, K. Hyams, J. Kaunda, B. Klaus, K. Michael, et al. 2017. Inserting rights and justice into urban resilience: a focus on everyday risk. Environment and Urbanization 29(1):123-138. https://doi.org/10.1177/0956247816686905

APPENDIX

Table 1: International and subnational policy documents consulted in review of framing of informality

Policy citation	**Type of policy**	**Description**
(Bongaarts 2019)	Global scientific assessment	This report represents a critical assessment, and the first ever carried out by an intergovernmental body, of the status and trends of the natural world, the social implications of these trends, their direct and indirect causes, and, importantly, the actions that can still be taken to ensure a better future for all. These complex links have been assessed using a simple, yet very inclusive framework that should resonate with a wide range of stakeholders, since it recognizes diverse world views, values and knowledge systems.
(IPCC 2022)	Scientific report based on IPCC Working Group II	This report recognizes the interdependence of climate, ecosystems and biodiversity, and human societies, and integrates knowledge more strongly across the natural, ecological, social and economic sciences than earlier IPCC assessments. The assessment of climate change impacts and risks as well as adaptation is set against concurrently unfolding non-climatic global trends e.g., biodiversity loss, overall unsustainable consumption of natural resources, land and ecosystem degradation, rapid urbanization, human demographic shifts, social and economic inequalities and a pandemic.

(UN-Habitat 2020)	Annual report	This Annual Report summarizes UN-Habitat's achievements in the first year of the Strategic Plan 2020–2023 which focuses on working in the most pressing areas of development such as poverty, prosperity, climate action and crisis prevention. This includes the full range of the continuum of human settlements from neighborhood communities to the largest urban conglomeration.
(Convention on Biological Diversity 2021)	Global biodiversity policy. Draft recommendation for Action Agenda. In December 2022 this draft was superseded the Kunming-Montreal Global Biodiversity.	This is a report on the Action Agenda for subnational governments and the report background, gives the objectives for the Action Agenda, it speaks to the activities to engage subnational government, cities and other local authorities and lastly, provides steps for the implementation of the Plan of Action
(Pörtner et al. 2021)	Scientific workshop report	The workshop report provided information relevant to the implementation of the Paris Agreement, the Post-2020 Global Biodiversity Framework and the Sustainable Development Goals.
(UN-Habitat 2016)	Summary report based on Habitat III- Quito 17-20 October 2016	The New Urban Agenda represents a shared vision for a better and more sustainable future. If well-planned and well-managed, urbanization can be a powerful tool for sustainable development for both developing and developed countries. The report provides a summary of the discussions/agenda points at the event hosted in Quito.
(Elmqvist et al. 2013)	Scientific Global assessment of urban biodiversity	Urbanization is a global phenomenon and the book emphasizes that this is not just a social-technological process. It is also a social-ecological process where

		cities are places for nature, and where cities also are dependent on, and have impacts on, the biosphere at different scales from local to global. This book is a global assessment of the above mentioned.
(McDonald et al. 2018)	Scientific global assessment with emphasis on climate impacts and conservation in urban settings. Convened by The Nature Conservancy	This report presents a business-as-usual scenario, which assumes that current urban growth trends will continue, and quantifies the impact that urban growth could have on biodiversity and human wellbeing. This report also quantifies the significance of natural habitat for climate mitigation and adaptation. It highlights solutions that can help avoid the negative impacts forecasted under our business-as-usual scenario, ways that governments at all levels can plan and implement a positive natural future for our urban century.
(Secretariat of the Convention on Biological Diversity 2012)	Policy brief, based on a global scientific assessment	The Cities and Biodiversity Outlook – Action and Policy provides a summary of a global assessment of the links between urbanization, biodiversity, and ecosystem services. Drawing on contributions from more than 120 scientists and policy-makers from around the world, it summarizes how urbanization affects biodiversity and ecosystem services and presents 10 key messages for strengthening conservation and sustainable use of natural resources in an urban context. It also showcases best practices and lessons learned, and provides information on how to incorporate the topics of biodiversity and ecosystem services into urban agendas and policies.
(Accra Metropolitan Assembly 2019, Addis Ababa	City Resilience Strategies: Accra, Addis Ababa, Dakar,	The six African city-level Resilience Strategies were developed under the 100 Resilient Cities program

City Administration 2020, Ville de Dakar 2016, Lagos State Government 2020, Ethekwini Municipality 2017, City of Cape Town 2019)	Lagos, Durban, Cape Town	(https://www.rockefellerfoundation.org/100-resilient-cities/) which was initiated by the Rockefeller Foundation in 2013. The intention behind the program was to catalyze a global urban resilience movement and to help the participating cities (which were selected through a competitive approach across the globe) to strengthen their urban resilience. As part of the program participating cities received resources and expert support to develop their context-specific resilience strategies. The strategies articulate the cities' resilience priorities and specific initiatives for short-, medium-, and long-term implementation. The main purpose of the resilience strategies is to trigger action, investment, and support within the city government and beyond.

LITERATURE CITED IN APPENDIX

Accra Metropolitan Assembly. 2019. Accra Resilience Strategy. Accra.

Addis Ababa City Administration. 2020. Addis Ababa Resilience Strategy. Addis Ababa.

Bongaarts, J. 2019. Summary for policymakers of the global assessment report on biodiversity and ecosystem services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services. Population and Development Review, 45, 680-681.

City of Cape Town. 2019. Cape Town Resilience Strategy. Cape Town.

Convention on Biological Diversity. 2021. Engagement with subnational governments, cities and other local authorities to enhance implementation of the post-2020 Global Biodiversity Framework. [online]: Convention on Biological Diversity.

Elmqvist, T., M. Fragkias, J. Goodness, B. Güneralp, P. J. Marcotullio, R. I. McDonald, S. Parnell, M. Schewenius, M. Sendstad, K. C. Seto and C. Wilkinson (eds.) 2013. Urbanization, Biodiversity and Ecosystem Services: Challenges and Opportunities. Dordrecht: Springer.

eThekwini Municipality. 2017. Durban Resilience Strategy 2017.

IPCC. 2022. Summary for Policymakers. In: H.-O. Pörtner, D. C. Roberts, E. S. Poloczanska, K. Mintenbeck, M. Tignor, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller and A. Okem (eds.) Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge, UK and New York, NY, USA: Cambridge University Press.

Lagos State Government. 2020. Lagos Resilience Strategy. Lagos.

McDonald, R., M. L. Colbert, M. Hamann, R. Simkin, B. Walsh, F. Ascensão, M. Barton, K. Crossman, M. Edgecomb, T. Elmqvist, A. Gonzalez, B. Güneralp, D. Haase, O. Hillel, K. Huang, D. Maddox, A. Mansur, J. Paque, H. Pereira and R. Sharp. 2018. Nature in the Urban Century: A global assessment of where and how to conserve nature for biodiversity and human wellbeing.

Pörtner, H. O., R. J. Scholes, J. Agard, E. Archer, A. Arneth, X. Bai, D. Barnes, M. Burrows, L. Chan, W. L. Cheung, S. Diamond, C. Donatti, C. Duarte, N. Eisenhauer, W. Foden, M. A. Gasalla, C. Handa, T. Hickler, O. Hoegh-Guldberg, K. Chii, U. Jacob, G. Insarov, W. Kiessling, P. Leadley, R. Leemans, L. Levin, M. Lim, S. Maharaj, S. Managi, P. A. Marquet, P. McElwee, G. Midgley, T. Oberdorff, D. Obura, E. Osman, R. Pandit, U. Pascual, A. P. F. Pires, A. Popp, V. Reyes-García, M. Sankaran, J. Settele, Y. J. Shin, D. W. Sintayehu, P. Smith, N. Steiner, B. Strassburg, R. Sukumar, C. Trisos, A. L. Val, J. Wu, E. Aldrian, C. Parmesan, R. Pichs-Madruga, D. C. Roberts, A. D. Rogers, S. Díaz, M. Fischer, S. Hashimoto, S. Lavorel, N. Wu and H. T. Ngo. 2021. IPBES-IPCC co-sponsored workshop report on biodiversity and climate change. IPBES and IPCC.

Secretariat of the Convention on Biological Diversity. 2012. Cities and Biodiversity Outlook. Montreal: Convention on Biological Diversity.

UN-Habitat. 2016. The New Urban Agenda: Habitat III. Quito: United Nations Conference on Housing and Sustainable Development.

UN-Habitat. 2020. Annual Report. Nairobi: United Nations Human Settlements Programme.

Ville de Dakar. 2016. Dakar Resilience Strategy. Dakar.